



Groundwater geochemistry using modified integrated water quality index (IWQI) and health indices with special emphasis on nitrates and heavy metals in southern parts of Tirupati, South India

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Abstract Groundwater is particularly vulnerable to pollution in places with a high population density and extensive human usage of the land, especially in southern parts of Tirupati, India. To assess this, 60 bore-well samples were obtained and assessed for physical specifications, ion chemistry, and heavy metals during the pre- and post-monsoon seasons 2022. The current investigation employed a modified integrated water quality index (IWQI), conventional graphical and human health risk assessment (HHRA) of nitrates and heavy metals to know the groundwater chemistry and its detrimental health effects on humans. The major ions were analyzed using American public health association (APHA) standards, whereas heavy metals were analyzed using inductively coupled plasma optical emission spectrometry (ICP-OES). Additionally, pH Redox Equilibrium and C (PHREEQC), a geochemical model written in C programming language was employed to determine the saturation indices of mineral facies and ArcGIS

10.3.1 was used for spatial distribution patterns of IWQI. Then, the HHRA of nitrates and heavy metals was performed using United States environmental protection agency (US EPA) guidelines. The noteworthy outcomes include elevated levels of Ca^{2+} , Mg^{2+} , Cl^- , NO_3^- , Cu, Fe, Mn, and Pb, demonstrating rock-water interaction, silicate weathering, Ca-Mg- HCO_3 followed by mixed water facies, dissolution/precipitation, reverse exchange, and anthropogenic contamination are the major controlling processes in groundwater of southern Tirupati, India. The modified IWQI reveals that most groundwater samples (38%) fall under the bad quality class, with (47%) in the poor quality class and only (15%) classified as medium quality class in pre- and post-monsoon seasons. Elevated IWQI were observed in all directions except in the east, which is suitable for drinking. Moreover, the major hazard quotient (HQ) and hazard index (HI) for nitrates (NO_3^-) and heavy metals like copper (Cu), iron (Fe), manganese (Mn), lead (Pb) and zinc (Zn) are above the critical value of 1, revealing potential risk to humans, especially infants, followed by children and adults, entailing the instantaneous implementation of proper remedial measures and stringent policies to reduce the risk associated with groundwater pollution in the southern parts of Tirupati.

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